PassFrame tuition

Fraction Operations

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 2 1/4 (9 ÷ 4 = 2 remainder 1)

2. B — 3 2/3 (11 ÷ 3 = 3 remainder 2)

3. 13/5 (2×5+3=13, denominator stays 5)

4. 2/6 + 1/6 = 3/6 (= 1/2)

5. A — 2/12 (= 1/6)

6. 2/5 ÷ 2 = 1/5 (divide the numerator by 2)

7. Whole numbers: 1+2=3; fractions: 1/4+1/4=2/4; total = 3 2/4 (= 3 1/2)

8. A, B and D (2 1/2 = 5/2; 10/4 = 5/2; 15/6 = 5/2; 2 2/5 = 12/5, which is not equal)

9. 3 2/5 (17 ÷ 5 = 3 remainder 2)

10. A — 1/7

11. Whole numbers: 2−1=1; fractions: 3/4−1/4=2/4; total = 1 2/4 (= 1 1/2)

12. 5/10 + 2/10 = 7/10

13. 4 cups (wholes: 1+2=3; fractions: 1/2+1/2=1 whole; 3+1=4)

14. B — 7/5

15. 1/2 × 1/4 = 1/8

Standard

1. Wholes: 1+2=3; fractions: 1/3=2/6, 2/6+1/6=3/6 (=1/2); total = 3 1/2

2. B — 1 1/2 (3 1/4 = 2 5/4; 2 5/4 − 1 3/4: wholes 2−1=1, fractions 5/4−3/4=2/4; total 1 2/4 = 1 1/2)

3. 2/3 × 3/5 = 6/15 (= 2/5)

4. 3/4 ÷ 3 = 1/4 (divide the numerator by 3)

5. B — 2/5 (4/5 ÷ 2 = 4/10, which simplifies to 2/5)

6. 10/12 − 3/12 = 7/12 (5/6 = 10/12, 1/4 = 3/12)

7. Wholes: 2+1=3; fractions: 1/2=2/4, 2/4+1/4=3/4; total = 3 3/4 litres

8. A, B and C (3/4 × 2/3 = 6/12, which equals 1/2 and also 2/4; D, 3/8, is not equal)

9. 4/10 + 3/10 = 7/10 (2/5 = 4/10)

10. B — 1/8

11. 4 1/5 = 3 6/5; 3 6/5 − 2 3/5: wholes 3−2=1, fractions 6/5−3/5=3/5; total = 1 3/5

12. 3/8 × 4/5 = 12/40 (= 3/10)

13. 7/8 = 21/24, 1/3 = 8/24; 21/24 − 8/24 = 13/24

14. A — 2 1/6 (1 1/2 = 1 3/6; 1 3/6 + 4/6 = 1 7/6 = 2 1/6)

15. 9/10 ÷ 3 = 3/10; 3/10 + 1/5 (2/10) = 5/10 (= 1/2)

Standard / Hard

1. Wholes: 2+1=3; fractions: 1/3=4/12, 3/4=9/12, sum=13/12=1 1/12; total = 3 + 1 1/12 = 4 1/12

2. A — 1 5/12 (3 1/4 = 2 15/12 after borrowing; 1 5/6 = 1 10/12; 2 15/12 − 1 10/12 = 1 5/12)

3. 29/6 = 4 5/6; 4 5/6 − 2 = 2 5/6

4. 4/9 × 3/8 = 12/72 (= 1/6)

6. C — 3 3/4 + 1/2 = 4 1/4, which is greater than 4 (A = 3 3/4, B = 3 5/6, D = 3, all less than 4)

7. 5/6 ÷ 5 = 1/6 per piece; 2 pieces joined = 2/6 (= 1/3) m

8. A, B and C (1/2×1/2 = 1/4; 3/4×1/3 = 3/12 = 1/4; 2/3×3/8 = 6/24 = 1/4; D, 1/8×3 = 3/8, is not 1/4)

9. 3 1/2 = 2 6/4 after borrowing; 2 6/4 − 1 3/4: wholes 2−1=1, fractions

11. 1 2/3, 1 3/4, 1 5/6 (as twelfths: 1 8/12, 1 9/12, 1 10/12)

12. 9/10 ÷ 3 = 3/10 kg

13. A, B and C (each pair differs by exactly 1/2; D, 2 1/4 and 1 1/2, differs by 3/4)

14. 2/5 × 3/4 = 6/20 (= 3/10)

6/4−3/4=3/4; total = 1 3/4 kg

5. 7/8 ÷ 7 = 1/8 m

10. A — 1/2 (5/6 ÷ 5 = 1/6; 1/6 × 3 = 3/6 = 1/2)

15. 5/8 × 1/4 = 5/32

Homework — Answers

Easy / Standard

1. 3 1/4 (13 ÷ 4 = 3 remainder 1)

2. B — 3 2/5 (17 ÷ 5 = 3 remainder 2)

3. 10/3 (3×3+1=10)

4. 2/8 + 1/8 = 3/8

5. A — 3/8

6. 4/5 ÷ 2 = 2/5

7. Wholes: 2+1=3; fractions: 1/3+1/3=2/3; total = 3 2/3

8. A, B and D (3 1/2 = 7/2; 14/4 = 7/2; 21/6 = 7/2; C, 3 1/4, is not equal)

9. 3 4/6 (= 3 2/3) (22 ÷ 6 = 3 remainder 4)

10. A — 1/8

11. Wholes: 3−1=2; fractions: 1/2=2/4, 2/4−1/4=1/4; total = 2 1/4

12. 4/12 + 3/12 = 7/12

13. Wholes: 1+2=3; fractions: 1/2=2/4, 2/4+1/4=3/4; total = 3 3/4 m

14. B — 4/7

15. 1/3 × 1/5 = 1/15

Standard

1. Wholes: 1+1=2; fractions: 2/5=6/15, 1/3=5/15, sum=11/15; total = 2 11/15

2. D — 0 4/5 (= 4/5) (2 1/5 = 1 6/5 after borrowing; 1 6/5 − 1 2/5: wholes 0, fractions 6/5−2/5=4/5)

3. 3/5 × 2/3 = 6/15 (= 2/5)

4. 7/8 ÷ 7 = 1/8

5. B — 1/6

6. 7/8 = 21/24, 1/3 = 8/24; 21/24 − 8/24 = 13/24

7. Wholes: 2−1=1; fractions: 1/2=2/4, 2/4−1/4=1/4; total = 1 1/4 m

8. A and B (3/5 × 5/6 = 15/30, which equals 1/2; C, 2/3, and D, 6/10 = 3/5, are not equal)

9. 6/8 + 5/8 = 11/8 (= 1 3/8)

10. B — 3/10

11. 4 1/6 = 3 7/6 after borrowing; 3 7/6 − 2 5/6: wholes 1, fractions 7/6−5/6=2/6 (=1/3); total = 1 1/3

12. 5/6 × 3/10 = 15/60 (= 1/4)

13. 2/3 = 10/15, 1/5 = 3/15; 10/15 + 3/15 = 13/15

14. A — 2 3/4 (1/2 = 2/4; 2 1/4 + 2/4 = 2 3/4)

15. 11/12 ÷ 11 = 1/12; 1/12 + 1/4 (3/12) = 4/12 (= 1/3)

Standard / Advanced

1. Double 1 1/4 = 2 2/4 (= 2 1/2); 2 1/2 − 1/2 = 2

2. 2 2/3 = 8/3; half = 4/3 (= 1 1/3); 1 1/3 + 1/3 = 1 2/3

3. B — 1 (1/6 + 1/3 = 1/6 + 2/6 = 3/6 = 1/2; doubled = 1)

4. Double 3/4 = 1 1/2; 2 1/2 − 1 1/2 = 1

6. 3 1/2 = 3 6/12; used: 1 2/3 = 1 8/12 and 3/4 = 9/12, total 2 17/12 = 3 5/12; left = 3 6/12 − 3 5/12 = 1/12

7. 2/3 = 10/15, 1/5 = 3/15; completed = 13/15; remaining = 2/15

8. A — 1/8 kg (3/4 ÷ 6 = 3/24 = 1/8)

9. Half of 7/8 used = 7/16, leaving 7/16; a further 1/8 (= 2/16) of the full

11. A and B (5 1/2 − 2 1/4 = 3 1/4 m, which is more than 3 m; the other piece is longer than the first piece, not shorter, and is not exactly half of 5 1/2)

12. 5/6 = 10/12; drained 1/4 (3/12) + 1/3 (4/12) = 7/12; left = 10/12 − 7/12 = 3/12 (= 1/4)

13. A — 3/4 kg (4 1/2 = 9/2; 9/2 ÷ 6 = 9/12 = 3/4)

14. 3/5 = 12/20, 1/4 = 5/20; raised = 17/20; still needed = 3/20

tank used; 7/16 − 2/16 = 5/16 left

5. A, B and C (double 1/3 = 2/3; 1/3+1/3 = 2/3; 1/3×2 = 2/3; only D, 2/6 = 1/3, is not equal)

10. 2/5 × 3 = 6/5; 6/5 ÷ 2 = 6/10 (= 3/5)

15. 1 3/4 = 1 9/12, 5/6 = 10/12; 1 9/12 − 10/12 = 11/12 (after borrowing); 11/12 × 2 = 22/12 (= 1 5/6)